Sound Analysis-Based Animal Recognition and Repel System for Agriculture *

Chamika M. Dissanayake
Department of Electrotechnology
Faculty of Technology
Wayamba University of Sri Lanka
Kuliyapitiya, Sri Lanka
Chamikauni@gmail.com

Priyashantha Tennakoon
Department of Electrotechnology
Faculty of Technology
Wayamba University of Sri Lanka
Kuliyapitiya, Sri Lanka
priyashanthat@wyb.ac.lk

2023 Moratuwa Engineering Research Conference (MERCon) | 979-8-3503-4521-6/23/$31.00 ©2023 IEEE | DOI: 10.1109/MERCon60487.2023.10355383

***Abstract*—Crop damage by animals and insects is a significant challenge faced by the agricultural sector. Recently, the number of crop damages due to animal interventions has increased. As a solution to this problem, an animal recognition and repel system is proposed. The proposed system detects the animal's presence, identifies the animal, and diverts the identified animal away from the agriculture field. A passive infrared (PIR) sensor detects the presence of any animal in the surrounding area and, the sound analysis system identifies the animal by analyzing the sound of the site. Further, the identified animal will be driven away by generating specific ultrasonic sounds that can only irritate the particular animal with a combination of a bright light-emitting diode (LED) that works only in a dark environment. Moreover, a message is transmitted to the owner by notifying the type of animal through a Global System for Mobile Communication (GSM) module. Therefore, the notified owner can work on additional protection methods. This paper presents the development and testing of the system, including its effectiveness in various environments and its ability to repel different types of animals. Moreover, this study provides valuable insights into developing and implementing animal-repellent systems for managing wildlife.**

***Keywords*—Animal repels, Agriculture, Sound analysis, Raspberry Pi**

I. Introduction

Sri Lanka has been an agricultural country since its immemorial. Exporting agricultural goods is one of the significant economic sectors of Sri Lanka. Coconut, tea, and spices are the three main types of agricultural goods that are currently been exported from Sri Lanka. The primary form of agriculture in Sri Lanka is rice production. Rice cultivation is a livelihood for more than 2 million Sri Lankan farmers [1]. The cultivation of vegetables is another major agricultural field in Sri Lanka. Out of the total population of Sri Lanka, 27.1 percent engage in agricultural activities [1].

The problem of lack of resources such as food, water, and land has arisen due to overpopulation. As a solution to solve this problem, people move to forest areas to find shelters and make a livable environment. Therefore, animals in the forest are stranded in small areas. This results, in unwanted visits of animals to the nearby villages for food. Animals such as elephants, monkeys, peacocks, wild boar are come into contact with humans. This collision between animals and humans has resulted in several negative impacts on both humans and animals. When animals enter the residential areas, they raid crop fields in search of food, causing the fields to be destroyed. Most of the farmers in rice production reported wildlife, including peafowl, wild boar, and free-roaming domestic animals such as buffaloes, were roaming around their sites and harmed their crop field [2]. Monkeys are another major problem for farmers in Sri Lanka, as they often raid fruit trees and vegetable gardens, causing damage to crops and reducing yields. Also, wild boars are causing significant issues, as they dig up crops and root out seedlings which cause widespread damage to farmlands. On the other hand, peafowl often feed on young seedlings, causing damage to rice paddies and vegetable gardens [2].

In the past, farmers used several techniques to repel animals from their farming fields. For example, placing a scarecrow in the field, spraying predator urine around the field, and use of chemical insecticides. Nowadays, farmers use electrical fences to protect their crop fields and living areas. Some animals, may harm the electric fence while approaching the field of crops and damage the crops. Also, electric fences cannot prevent the entrance of small animals into the crop field. Therefore, the animal-repelling system will be a huge help to the farmers to protect their crops from these predators.

Most animals have unique behavior to them. Unlike predators, Wild creatures such as monkeys, deer, peacocks, and elephants hunt for food at specific times that they choose based on their habits. It has been confirmed that the simple and most common method for animal repelling is emit bright light toward the animal [3]. This method decreased animal activity, and they strive to hide. The research has confirmed that using a flashlight to repel animals has better results compared to other conventional methods. In [4], ultrasonic sound has been used to repel rats to confirm that the use of ultrasonic sound has a good impact on repelling animals. The ultrasonic sound is tested in a laboratory environment to check the response of rats. Animals respond to different ultrasonic frequencies depending on their hearing range of the animal. Three frequency levels are identified to repel different animals [5]. Table 1 gives the details of frequencies that irritate animals with the animal type.

979-8-3503-4521-6/23/$31.00 ©2023 IEEE

TABLE I. IRRITATING SOUND FREQUENCY FOR ANIMAL

Animals	Frequency range
Birds	15 kHz – 30 kHz
Dogs	20 kHz – 25 kHz
Cat	23 kHz – 27 kHz
Monkeys	13 kHz – 16 kHz
Squirrels, Deer, peacock	10 kHz – 35 kHz
wild boar, buffalo	36 kHz – 65 kHz
Rat, Mouse	20 kHz – 60 kHz

Several animal repel devices have been developed in the state of the art of research. An animal repel system has been proposed in [6]. This device has been designed by using an Arduino UNO microcontroller working with an IR sensor, ultrasonic sensor, and a GSM module. Infrared radiation (IR) sensors will placed around an agricultural field and when the sensors detect an animal, an ultrasonic sound will played to divert the animal. However, this system does not identify the animal and emits an ultrasonic sound with the same frequency irrespective of the animal type. In [7], a smart ultrasonic animal and insect rappeler through Internet of Things(IoT) has been designed. This device consists of an ATmega16 micro-controller board, an IR Sensor, a piezo-electric buzzer, and a Direct Current(DC) motor. The device used the IR sensor for animal detection and consequently turns on the piezo-electric buzzer to generate a sound to repel the detected animal. However, the buzzer can not produce a high-frequency sound. The IR sensor will triggered every time it detects any movement in the surrounding area of the sensor. This generate many false alarms resulting malfunction of the device.

In [8], an Arduino-based animal repel device has been proposed. Ultrasonic sensors of this device detect the animal present in the area and emit a sound at three distinct frequencies to frighten away three different kinds of animals. For one group, several animals are included. The sound output of the system is 50 kilohertz (kHz) for the first 2 seconds, followed by 32 kHz for the following 4 seconds, and 18 kHz for the last 6 seconds. This device is triggered when the ultrasonic sensor detects any movement of the sensor area. However, this device is limited to three different frequencies and repels only three different group types of animals. A real-time animal repellent system using image processing has been developed in [9]. In this system, a surveillance camera around the field will record 2-minute videos and transfer them via radio-frequency (RF) transmitter to a computer where the image is stored and analyzed with pre-stored data in a database. An algorithm has been developed to detect the animals by comparing data from recorded video to the pre-stored data. When the developed algorithm detects an animal, the alarm will triggered and ultrasonic sound waves between 16–25 kHz are played to repel all kinds of animals. This system requires a high-performance processor and it needs to be active at all times when the animal repels system is used. Also, the accuracy of the device is low because the result of animal recognition needs to be transferred to the device via an RF transmitter.

In a better animal-repelling system, identification of a greater variety of animal species is important. Also, accurately identifying the type of animal, and emitting a larger range of ultrasonic sounds is also essential. Additionally, it must be capable of alerting the farmer with a message. To this end, we developed an automated animal detection and repelling system. The key contribution of this work can be summarized as follows.

1) An animal detection system is developed. This detection system uses a passive infrared (PIR) sensor to detect the movement of any animals in the coverage area of the sensor.
2) An animal identification system with a sound analysis algorithm is developed to recognize the type of animal.
3) An animal repel system is developed. This system emits an irritating ultrasonic sound and blinks a light-emitting diode (LED) flashlight to scare animals.
4) The developed device uses the global system for mobile communication (GSM) module to alert the farmer with a warning text message.

The rest of the paper is organized as follows: The proposed animal repel system is presented in Section II along with the details of the animal identification system. In Section III, the animal repel system is provided and the animal repel methods and user alert system are explained. Testing of the device is shown along with a discussion of the results obtained by the survey in Section IV. Finally, Section V presents concluding remarks.

II. Animal recognition system

The proposed animal recognition system includes two PIR sensors, a sound sensor (microphone), and a light-dependent resistor (LDR) based light sensor. These sensors are connected to a processor (Raspberry Pi model 4). A battery is used to power up the system and a battery charger with a solar panel is used to charge the battery. This power-up system made possible to use the device as a portable device. Sensor data processing and controlling of the recognition system is performed by the processor. Also, the same processor is used as the processing element of the animal repel system. A speaker, an LED flashlight, and a GSM module are used as output devices of the animal repel system. Fig. 1 shows the block diagram of the animal recognition and repels system. The animal recognition system performs two functions which are animal detection and animal identification.

A. Animal Detection System

The device uses two HC-SR501 PIR sensors to detect the presence of an animal. With the use of two sensors, the device coverage area of detection is expanded. Two sensors can be mounted facing any direction as per the preference of the user. The detection system operates continuously for 24 hours to detect any movement of animals within the coverage areas of the PIR sensors. When the animal detection system is triggered, the animal identification system is initialized.

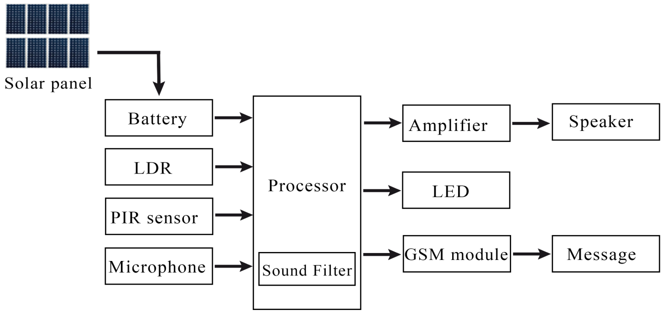

Fig. 1. Block diagram of animal recognition and repel system.

B. Animal Identification System

The animals are identified by analyzing the sound of the coverage area of the PIR sensors. The microphone is used to record the sound from the coverage area and pass it to the processor. A software filter is used for the noise reduction of incoming sound waves. The filtered sound wave is forwarded to the sound analysis system to identify the type of animal in the coverage area. In the sound analysis system, the received sound is compared to a trained model. This model has been developed using a sound pattern recognition system called Short-term Sound Pattern Recognition System (SOPARE) [10]. The real-time sound input from the microphone is fed into the trained model and consequently, the model detects the patterns of sound based on superficial characteristics. Afterward, The model identifies the relevant animal and gives as its output.

Algorithm 1: Sound Analyse algorithm

Data: Sequence of the sound pattern of animals
Result: Return the name of raw file
initialization;
Train the sound to the program;
compile and store sound in audio factory;
initialize configurations(set threshold value);
Begin;
while do
    Recorder `/* record the sound */`
    Buffering ;
    Processing `/* transform the sound into characteristics to be analyzed */`
    Filter;
    while *Guess the Best match* **do**
        Open the audio factory;
        if *Recorder.raw == Audio[num].raw* **then**
            Output the name of the raw file;
        else
            Open the following audio.raw file;
        end
    end
end

The basic process of the sound analysis algorithm is given in algorithm 1. The sound analysis program initializes with the default configuration settings given by the developer. This default configuration includes the sample rate of the sampling audio, with the chunk rate and noise canceling threshold value. The values indicate the quality of the sound samples of the model and the quality of the real-time recording sound. The initialization of SOPARE includes the configuration of the algorithm with the audio factory. The program with the audio unit testing algorithm and the audio configuration algorithm is included in the audio factory. Initially, the algorithm listens to the surrounding sound patterns and records a chunk of data. The recorded data chunk is filtered, processed, and transformed into chain characteristics to be analyzed by the algorithm. The chain of characteristics is compared with the trained result of characteristics, and the result of plugins is called the result.

C. Noise cancellation Method

The sound recorded by the microphone consists of the required sound pattern along with the noisy sound as depicted in Fig. 3. The required sound has a higher amplitude when compared to the noise of the environment. The amplitude of the noise remains in the same range of values. The noise is removed by analyzing the amplitude of the noise and adding a threshold value to the noise amplitude. The sound input above the threshold value is applied to the sound pattern recognizer for further processing.

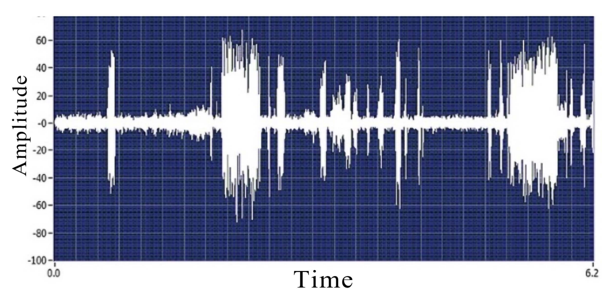

Fig. 2. Input Sound Spectrum.

III. Animal Repel system

The animal repelling system repels the animals from the crop field. The animal recognition system provides the recognized animal type to the repel system, then the animal repels system is activated. The repel system uses two methods to repel the animal from the crop field, 1) lightning-bright light, and 2) emitting an irritating ultrasonic sound that is specific to the identified animal. Also, this system has a user alert subsystem that sends the owner a message about the animal attack on the crop field.

A schematic diagram of animal detection and animal repel systems is shown in Fig. 2. As depicted in the figure, the main power unit of the circuit is a Li-ion battery pack that is connected to the TP4056 charging module. The charging

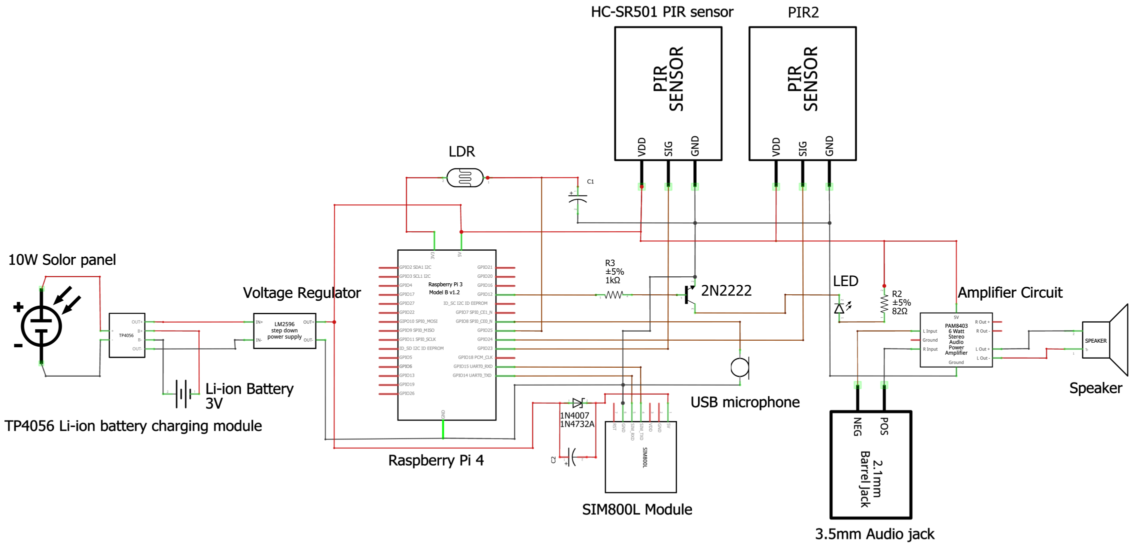

Fig. 3. Schematic diagram of animal detection and repel system

module gets the power from the connected solar panel and supplies the required voltage to the battery. The operating voltage of the processor is 5V, and a constant 5V is obtained by the battery using a voltage regulator. Two PIR sensors are connected to General Purpose Input/Output (GPIO) pins 23 and 24 of the processor. SIM800L GSM module needs a 3.8V - 4.2V power supply to power up. Therefore, 5V power is connected to the GSM module through a 1N5400 diode. The 1N5400 diode has a forward current of 3A, and the diode forward voltage is in the range of 0.8 – 1.0V. The diode acts as a voltage regulator to supply the required 3.8V - 4.2V to the GSM module. If the supply current is higher than 3A, There can be sparks in the GSM module. The diode acts as a current limiting element to prevent the flow of over-current to the GSM module. Receiving (RX) and the Transmitting (TX) pins of the GSM module are connected to the TX and RX pins in the processor. Also, the LDR is connected to GPIO pin 25, and the LED is connected to GPIO pin 12. Moreover, the ultrasonic sound is output by using a speaker connected to the 3.5mm audio jack of the processor.

A. *Animal Repel Methods*

- Lightning Bright Light

Naturally, bright light can be an irritation to the eyes of animals. Animals are afraid of suddenly switched-on bright lights in dark environments. This system utilized the aforementioned phenomenon as a repel method by emitting a blinking bright light [3]. The bright light repels method can only be used in dark time periods. By using a light sensor, the darkness of the environment is detected and with the use of power transistors high luminaries LED light is switched on if a relevant animal is detected.

- Emitting Noisy Ultrasonic sound

Bright light repel system is useful only when the environment is darker than the system set limit. However, a sound-based repel system can be used anytime to repel animals and is also a very efficient method. Ultrasonic sound is irritating to animals in some frequency ranges. The animal ear is more sensitive to high-frequency ultrasonic sound and these animals cannot withstand high-frequency ultrasonic sound for a long time period. With the use of new technology, any sound can be generated with different frequencies. The processor generates high-frequency ultrasonic sound and by using speakers of the device, this sound will be emitted to the environment. Here, the frequency of the ultrasonic sound will be selected according to the recognized animal by the animal recognition system.

- User Alert system

A warning message is delivered to the customer/owner of the crop field when an animal is detected on the coverage area of the sensor and after identified by the device. The main purpose of this system is to notify the owner about the animal attack for further necessary actions. The processor generates a text message with relevant information and then the system uses the SIM800L GSM module to send the warning text message to the owner.

B. *Survey*

In the development of the animal identification and repelling system, it is important to know about the type of harmful animals to the crop cultivation, the type of crop that gets damaged most from the animal attacks, and animal-repelling techniques already employed by farmers to repel these animals. To this end, a survey is conducted to collect the aforementioned information from farmers.

Data was collected by a physically conducted questionnaire survey. Candidates were selected from the villages around

Matale district Sri Lanka and farmers who cultivate paddy, potatoes, beans, bananas, cassava, and brinjal contributed to the questionnaire survey. In this survey, the eldest farmer in the family was interviewed. An analysis of the data collected from this survey is presented in Section IV.

IV. Result and Discussion

With the survey, data has been collected from different questions. The frequency analysis of Statistical Package for the Social Sciences (SPSS) software is used to identify the frequency of a response that has been collected from the survey. With this analysis of the selected data, the most given answers can be determined. With the result provided by analysis, the type of harmful animals to the crop field, the most suitable repelling method currently used by farmers, and the crop type that gets the most damage by animals are identified. According to the questionnaire results, rats, peacocks, and wild boars are the most harmful animal types for cultivation. The frequency of damage by monkeys is low, but the harmfulness from the monkeys is high. Fig. 4 shows the frequency of animal damage to the crop field.

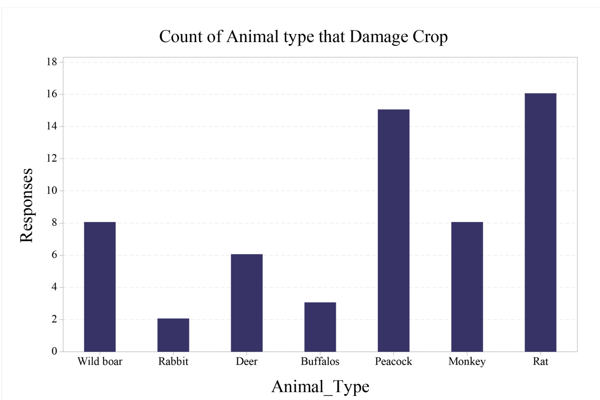

Fig. 4. Frequency of animal type.

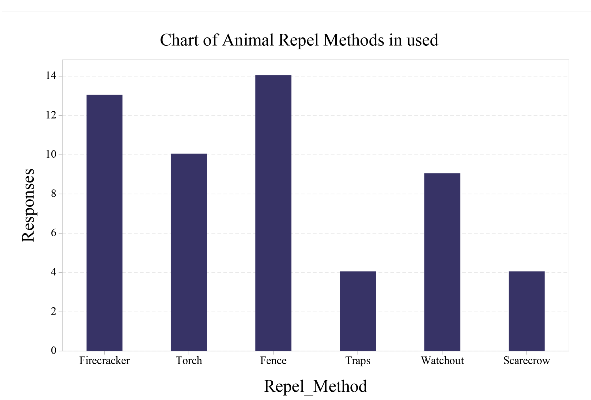

Fig. 5. Frequency of use animal repel methods.

TABLE II. TESTING RESULTS OF ANIMAL RECOGNITION SYSTEM

Successful attempt count						
Distance	**Cow**	**Peacock**	**Monkey**	**Dog**	**Total**	**Accuracy**
1.5 m	8/10	6/10	5/10	5/10	24	60%
3.0 m	10/10	7/10	6/10	8/10	31	77.5%
4.5 m	10/10	7/10	8/10	9/10	34	85.5%
6.0 m	8/10	9/10	7/10	9/10	33	82.5%
7.5 m	8/10	9/10	6/10	8/10	31	77.5%
9.0 m	8/10	8/10	6/10	7/10	29	72.5%

The majority of farmers use fences to keep animals away from their crops. More frequently used types are wire fences and cover nets. Also, most farmers guard their cultivation fields at night and use firecrackers and torches to repel animals. The frequency of used animal repelling methods by farmers is shown in Fig. 5. According to the survey, 23 farmers out of 30, do their cultivation near to the forests. With the high demand for space for cultivation, they selected areas near to the forests to cultivate. Farmers lose their harvest every year due to animal damage. As per the analysis, the majority of the farmers lose approximately 25 percent harvest due to the above reason. There were only two farmers, out of 30, who lost their harvest by approximately 75 percent. With the help of this frequency analysis, it was discovered that the cultivated fields closest to the forest sustain the most animal damage.

With the above analysis of the data obtained from the survey, it can be concluded that the most suitable way to repel animals is to use bright blinking lights and noisy ultrasonic sounds for animal repelling. The ultrasonic frequency is created from third-party software, and the sound file is added to the processor to playback the sound. The Audacity software is used to generate the required ultrasonic sound file.

To test the efficiency of the animal recognition system, the implemented device was tested around the nearest cultivation fields for 4 different animal types. The testing procedure was done by varying the distance between the animal and the device to detect the most accurate working distance. 10 different animal was tested in each animal type and the successful attempts count is included in Table II. Fig 6 shows the graphical representation of the testing results. Testing results prove that the animal recognition system has more accurate results when the animal is in a 6-meter radius area. Fig. 7 shows the implementation of animal recognition and animal repel systems along with the components connected to the device. The testing procedure of the device is shown in Fig. 8.

Developed animal identification and repels device has more advantages than existing animal repeller devices in the literature. The most useful feature is the recognition of an animal by sound analysis and the use of ultrasonic sounds with a specific frequency for the detected animal. With this special feature, the device has more accuracy than existing devices in repelling a wide range of animals. Also, the user warning system is an advantage in the developed system. This alerts the farmers about the animal intervention and they can take necessary actions for further protection of crop fields.

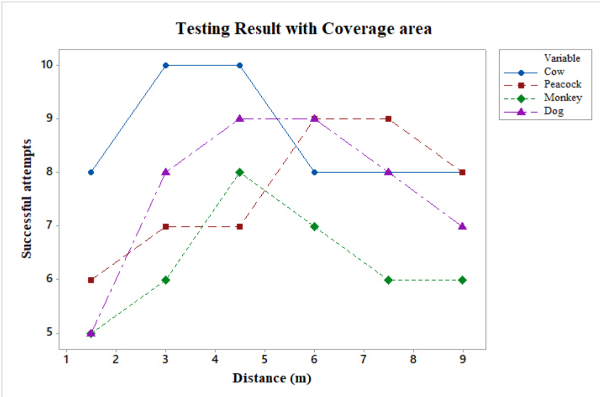

Fig. 6. Testing results of animal recognition system

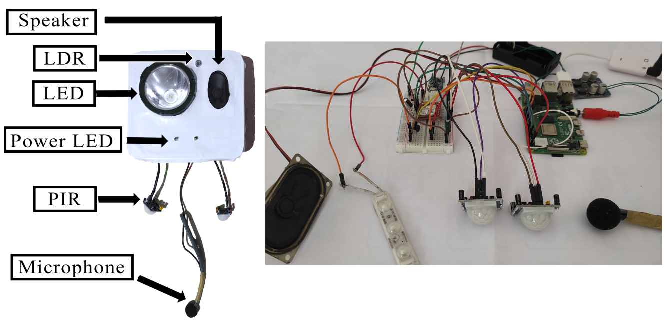

Fig. 7. Implementing of animal recognition and repel device

Fig. 8. Testing of animal recognition and repel device

V. Conclusions

An animal recognition and repel system has been developed. The developed system has an animal detection system, animal identification system, and animal repelling system. A PIR sensor has been utilized to detect the presence of the animal in the sensor's surrounding area. A sound sensor has been used to obtain the sound from the environment and it has been used to identify the arrived animal. A sound analysis algorithm has been developed to identify the animal from the obtained sound. A high-performance speaker has been used in the repel system to output ultrasonic sound with a specific frequency to the identified animal to repel the animal from the field. Bright Light system has been used as an additional repelling method. The developed system has advantages compared to the existing systems such as identifying an animal by using a sound analysis algorithm, emitting ultrasonic sound with a specific frequency to the identified animal, and sending a warning message to the farmers. The developed device was able to divert animals from the crop field.

References

[1] Department of Census and Statistics.(Nov. 01, 2021). *Sri Lanka labour force statistics quarterly bulletin*[online].Available: www.statistics.gov.lk.

[2] F. G. Horgan, E. P. Kudavidanage.“Farming on edge: Farmer training to mitigate human-wildlife conflict at an agricultural frontier in south Sri Lanka, Crop Protection”, Volume 127,2020,104981, ISSN 0261-2194

[3] M.L.F.Fonck, M.Gatta, C.Musyoki, M.V. Zelfde,G.A.Persoon, K.C.Musters, G.R.de Snoo, and H.H.de Iongh. “Effectiveness of a LED flashlight technique in reducing livestock depredation by lions (Panthera leo) around Nairobi National Park”. PLoS ONE 13(1).e0190898.

[4] S. M. Brudzynski and E. M. Chiu, “Behavioural Responses of Laboratory Rats to Playback of 22 kHz Ultrasonic Calls,” *Physiology and Behavior*, Vol. 57, No. 6, 1995, pp. 1039-1044.

[5] J. D. Sonone , D. A. Patil, K. P. Rane. “Irritating and Hearing Frequency Identification and Generation to Avoid Animals Accident”. *International Journal of Innovative Research in Science, Engineering, and Technology*. Vol. 3, Issue 7, July 2014

[6] M. J. Prabha, R. Ramprabha, V. V. Brindha, C. A. Beaula, “Smart crop protection system from animals”. *International Journal of Engineering and Advanced Technology (IJEAT)*. 9 pp. 2064-2067 (4), April 2020, ISSN: 2249 – 8958

[7] A. Ghoshi, B. Garg, “Smart Ultrasonic Animal and Insect Repeller Through IOT” *International Journal of innovative research in technology (IJIRT)* Volume 7 Issue 4 ISSN: 2349-6002. September 2020

[8] P. Sreevardhan, B. Raju, and D. Nandan, “Animal Repellents from Agricultural Fields”. *Next Generation Information Processing System, Proceedings of ICCET 2020*, Volume 2 (pp.109-121)

[9] S. Jalaja, A. Daniel, B.G. Shankar, M. Kaliraj, J.Y. Khanna, “Real-Time Animal Repellent System using Image Processing”. *International journal for acientific research and development (IJSRD)* Vol. 4, Issue 02, 2016 ISSN: 2321-0613

[10] Anon.(Jan. 11, 2022). *Smarthome, speech recognition, voice control, Raspberry Pi*.Available:https://www.bishoph.org/ .

[11] L. Tacioli, L. Toledo, and C. Medeiros. ”An Architecture for Animal Sound Identification based on Multiple Feature Extraction and Classification Algorithms”. *Proceedings of the XI Brazilian e-Science Workshop*. 2017, pp. 29-36

[12] C.Y.Yeo, S.A.R.Al-Haddad and C.K.Ng, “Animal voice recognition for identification (ID) detection system,”. *2011 IEEE 7th International Colloquium on Signal Processing and its Applications*. 2011, pp. 198-201

[13] J. Redij, P. Shitap, S. Singh, D. Zagade, S. Chougule, “Smart Crop Protection System from Animals.” *International Journal of creative research thoughts (2022)*.Volume 10, Issue 4 April 2022 — ISSN: 2320-2882

[14] P.U.Maheswari and A.R.Rajan.”Animal intrusion detection system using wireless sensor networks”. *International Journal of Advanced Research in Biology Engineering Science and Technology (IJARBEST)*. Vol. 2, Special Issue 10, March 2016. ISSN 2395-695X

[15] A. A. Shetgaonkar and V. N. S. Principal, “Smart animal repelled”. *International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS)*, 2017, pp. 3303-3308